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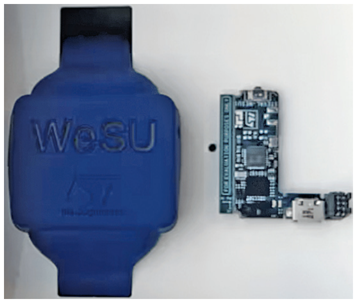
Wearable Demo	Wearable Sensor Unit	Temperature Sensor for Wearable Devices
		
The wearable demo design features ultra-low power Bluetooth, flexible motion and environmental sensors, a powerful microcontroller, and cryptographic operations, providing a compact, efficient solution for wearables.	The wearable reference design offers real-time monitoring, advanced sensors, BLE connectivity, a powerful microcontroller, comprehensive power management, and includes a watch strap for immediate use.	The temperature sensor reference design for wearables features a highly accurate sensor with rapid thermal response and a USB form-factor PCB with breakout tabs for versatile substrate attachment.
The wearable demo reference design is powered by key components from Atmel. Central to this platform is the ATBTLC1000, an ultra-low power Bluetooth SMART System on a Chip that integrates a Cortex-M0 MCU, transceiver, modem, MAC, PA, TR switch, and power management unit. The chip can function either as a Bluetooth low energy link controller or data pump with an external host MCU, or as a standalone application processor with embedded BLE connectivity and external memory. The platform's core sensors include the BHI160, which integrates a 6-axis IMU with the Bosch Sensortec Fuser core for flexible, low-power motion sensing and sensor data processing, and the BME280, which combines digital humidity, pressure, and temperature sensing with high accuracy and fast response times. The VEML6080 is an advanced ultraviolet and ambient light sensor. The design features the SAML21G18B processor and an array of sensors, including the BHI160 motion sensors and the BME280 environmental sensors. The platform includes the ATECC508A for cryptographic operations and LED indicators for operational status display. The device is CE/FCC certified and measures 40mm x 30mm, excluding the programming header extension.	The STEVAL-WESU1 from STMicroelectronics is a reference design optimised for wearable and portable applications, fitting compactly within a watch strap for real-time activity monitoring and sensor data acquisition. The design includes an application layer based on the BlueST protocol for data streaming from various sensors and devices, and a serial console over BLE for configuration adjustments. Key components include the STM32L151VEY6 microcontroller, LSM6DS3 accelerometer and gyroscope, LIS3MDL magnetometer, LPS25HB pressure sensor, and BlueNRG-MS Bluetooth processor. Connectivity is enhanced by a BALF-NRG-01D3 balun, and power management is handled by the STNS01 Li-ion charger and STC3115 fuel gauge. Powered by a 100mAh Li-ion battery, the device features a micro USB connector for recharging, an SWD connector for debugging, and includes a watch strap with plastic housing for immediate use. It is FCC and IC certified, meets RoHS standards, and is tested to UN38.3 for safety and environmental compliance.	TIDA-00452 is a temperature sensor reference design from Texas Instruments for wearable devices, featuring the LMT70 sensor with 0.13°C accuracy at human body temperatures. Its compact WCSP package allows quick heating and rapid thermal response. The design includes a USB form-factor PCB board with breakout tabs for attaching various substrates, detailed in a design guide that covers thermal response and calibration techniques for the MSP430F5528 ADC. Suitable for industrial and personal electronics, it includes an application processor module, electronic thermometer, probe board, and wearable fitness monitor. In personal electronics, it features a dashboard camera, GPS navigation device, smart trackers, and smartwatch. The LMT70, an analogue output temperature sensor with high accuracy and an output enable switch, detects temperatures from -55°C to 150°C. The LMT70EVM records the LMT70's output voltage across substrates, with a USB form factor board breakout tab for remote measurements, equipped with an MSP430F5548 microcontroller to capture output voltage through its integrated ADC.
The reference design is suitable for applications requiring rapid context awareness and maintains high overall accuracy across a wide temperature range.	The reference design is suitable for wearable motion sensing applications.	The reference design is suitable for both industrial and personal electronics applications.
Atmel	STMicroelectronics	Texas Instruments (TI)
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